A redox-active polymeric network facilitates electrified reactive-capture electrosynthesis to multi-carbon products from dilute CO_2-containing streams

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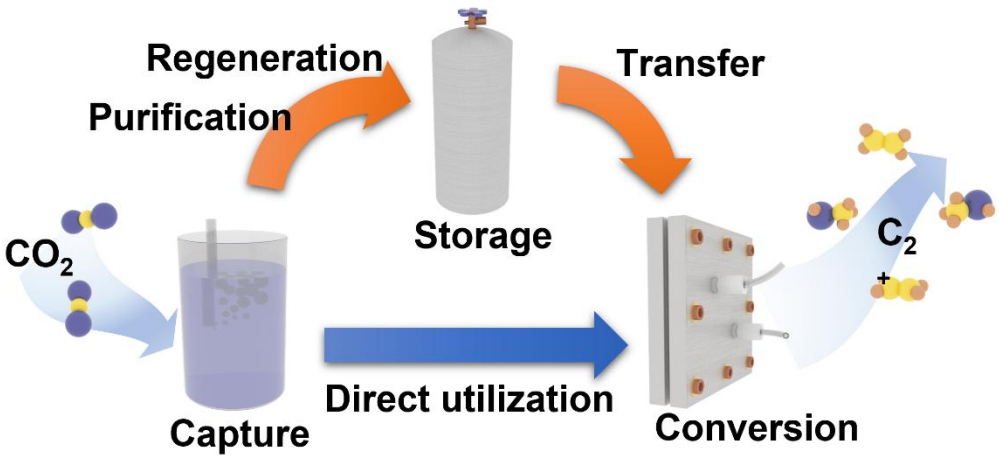

Figure S1 | Schematic illustration of the direct utilization of CO_2-capture liquids which detours around the extra energy input. The conventional electrochemical CO_2 reduction requires pure CO_2 as the carbon source, which induces additional energy input of CO_2 regeneration, purification, storage and transport. The direct utilization of CO_2-capture liquids can detour around these extra energy inputs by feeding these liquids as carbon sources instead of pure CO_2. Furthermore, there is often no CO_2 in the outlet of direct utilization, further bringing down the cost of product separation and purification.

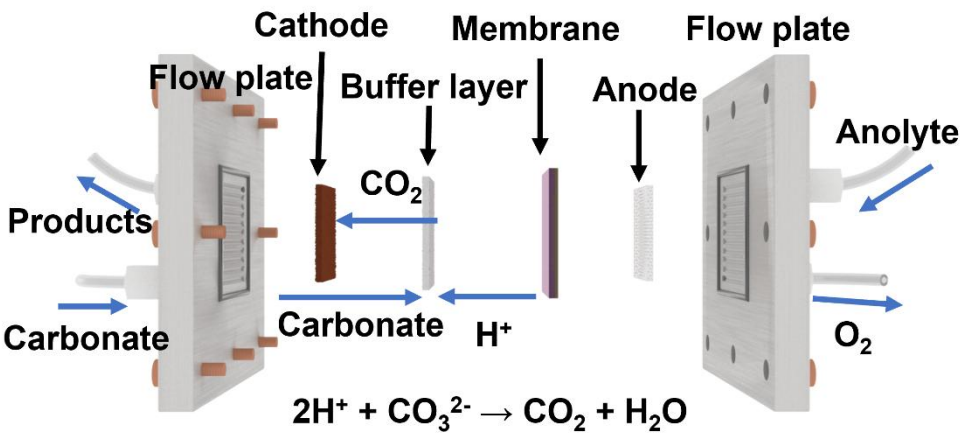

Figure S2 | Schematic illustration of the structure of the electrochemical cell which converts carbonate solution in C_{2+} products by in-situ generating CO_2.

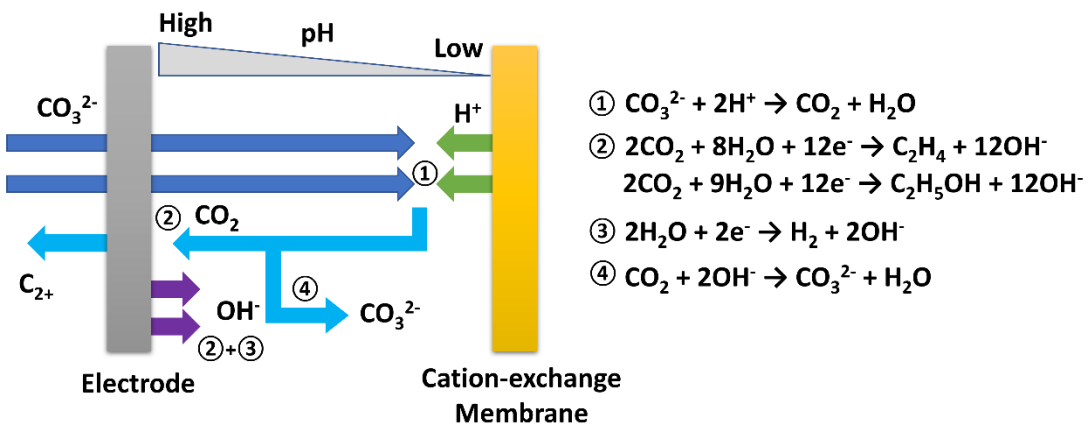

Figure S3 | Schematic illustration of the mechanism of carbonate reduction at the catholyte side. The carbonate reduction system relies on the in-situ generation of CO_2 from the reaction between protons and carbonate ions. The protons come from the cation-exchange membrane (CEM) or bipolar membrane (BPM) (Equation 1). As the amount of protons transferred from the membrane is limited by current density and the concentration of carbonate is not high, the in-situ generated CO_2 is usually lean, and a high proportion will be consumed by the hydroxides generated on the electrode when the CO_2 reduction occurs (Equation 2-4).

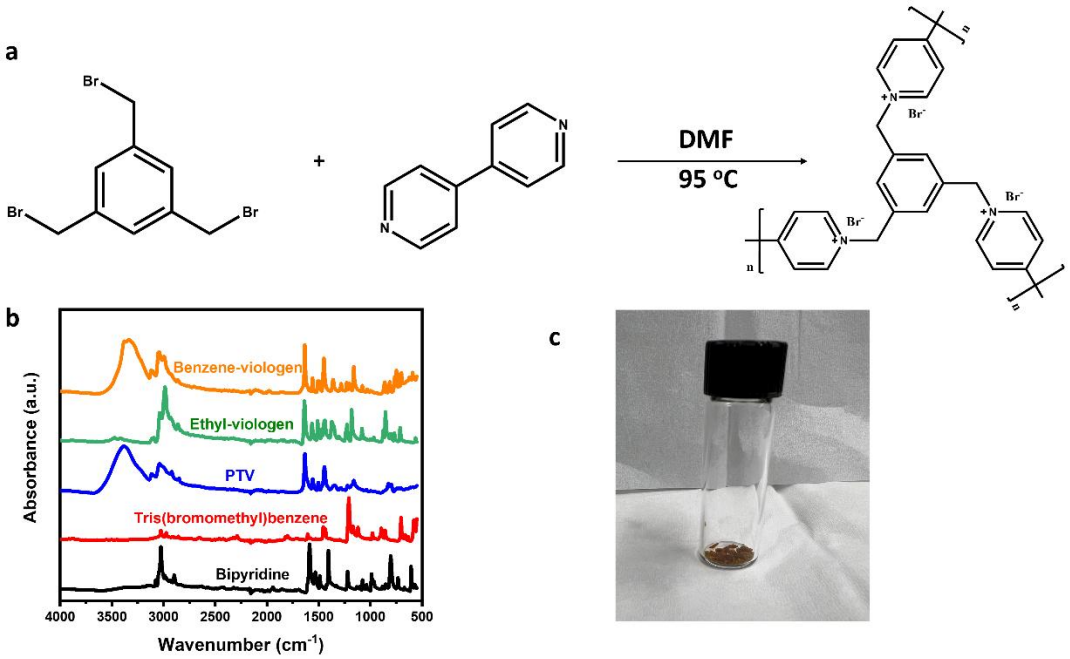

Figure S4 | Synthesis and characterization of PTV. a. The schematic illustration of the PTV synthesis. **b.** The FTIR spectra of the as-synthesized PTV and precursors. **c.** The digital photo of the as-synthesized PTV.

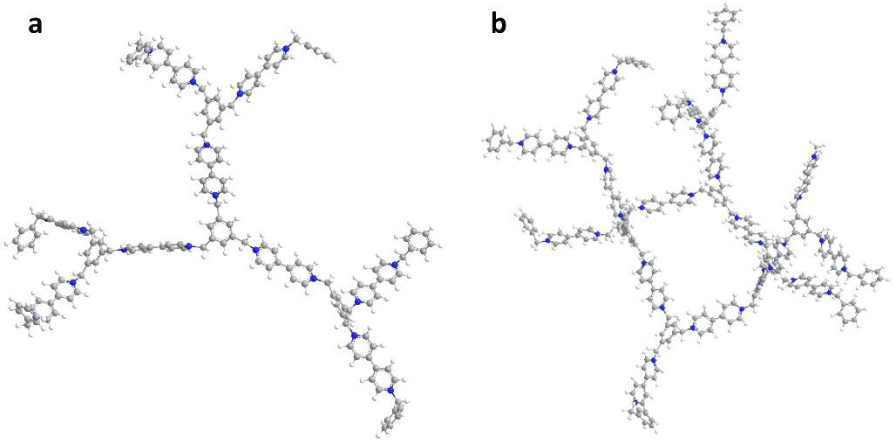

Figure S5 | The simulated 3d structure of the PTV fragments with a. n=1 and b. n=2 in the molecular formula.

By introducing the non-conjugated sp^3 hybridized methylene groups ($-CH_2-$) to bridge the conjugated viologen branches and benzene nodes, the overall molecular structure of the PTV polymer does not present a 2D conjugated structure. Instead, it will form a twisted network owing to the steric hindrance (Figure S5). Furthermore, the conjugated network can also help to maintain its mechanical integrity, which will generate molecular void in the twisted 3D network. The subsequent stacking of the PTV polymeric network will generate molecular pores which will be beneficial for CO_2 adsorption and trapping.

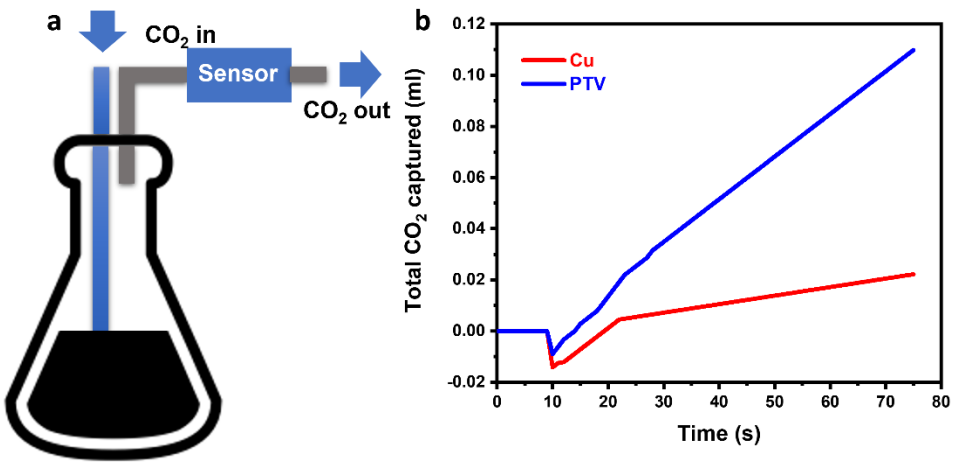

Figure S6 | a. The illustration of the demonstration experiment of CO_2 trapping and b. the total CO_2 captured with time.

In order to demonstrate the capability of PTV in attracting and trapping CO_2, we performed the demonstration experiment by connecting a CO_2 detector at the outlet of a flow bottle with the materials (Figure S6a). The bottle was firstly purged with N_2 and then CO_2 was slowly purged in the bottle. The CO_2 detector detected the flow rate of CO_2 during this process. As shown in Figure S6b, the CO_2 captured in PTV significantly increased compared to that in Cu, demonstrating that PTV has the capability in attracting and trapping CO_2.

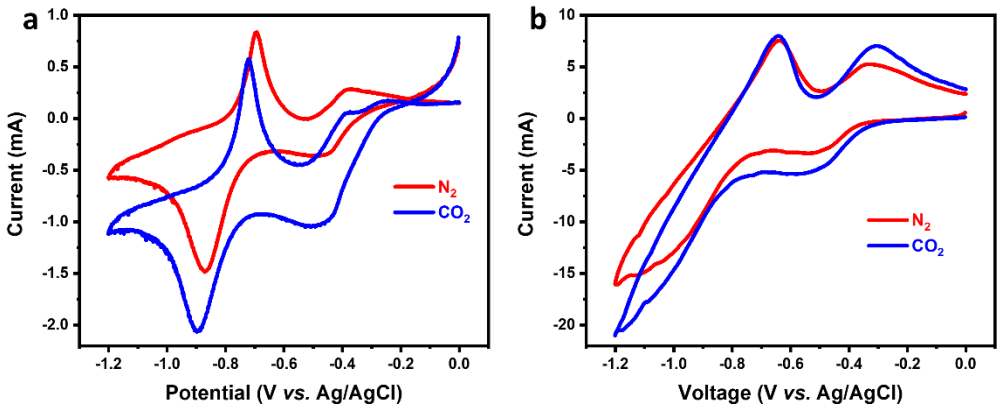

Figure S7 | Cyclic voltammetry curves of PTV electrodes in N_2 and CO_2 atmospheres at a potential range from 0 to -1.2 V with scan rates of a. 5 mV s^{-1} and b. 500 mV s^{-1}. Both CV curves show the increase of the reduction peaks after the purge of CO_2 in the system, indicating that the reductive form of PTV has the capability to activate CO_2 molecules by transferring electrons to CO_2.

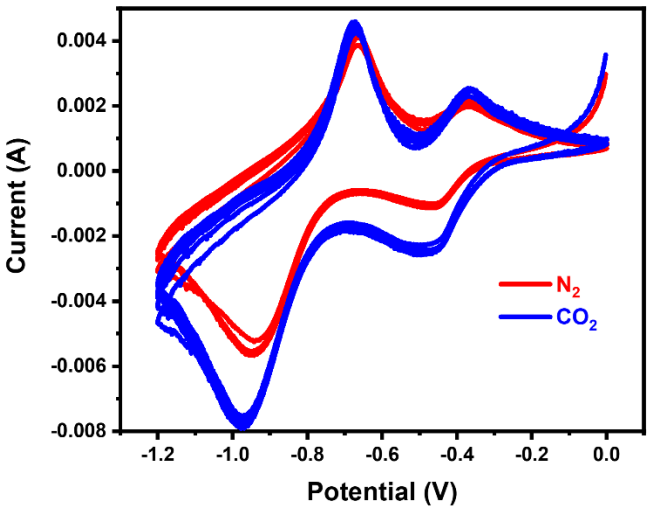

Figure S8 | Cyclic voltammetry curves of PTV electrode in N_2 and CO_2 atmospheres under repeated scans.

The different scans show identical curves, indicating that both the redox reactions of PTV in the N_2 atmosphere and the activation of CO_2 molecules in the CO_2 atmosphere are highly reversible.

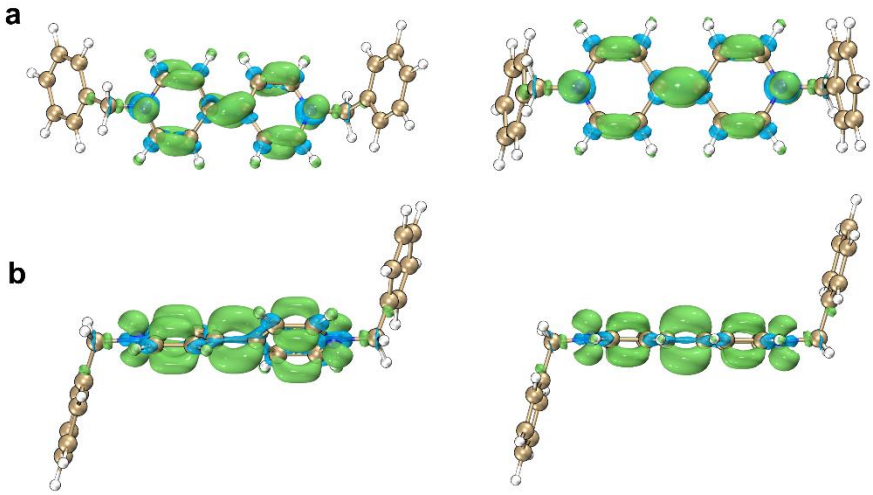

Figure S9 | The Fukui function of a. PV^{2+} and b. PV^{+} from the DFT calculations. The green and blue isosurfaces represent the positive and negative regions of the Fukui function, respectively. The positive regions indicate areas of high electrophilic character. The results show that both PV^{2+} and PV^{+} have strong electron-accepting abilities, which aligns with the observed continuous decrease in potential energy.

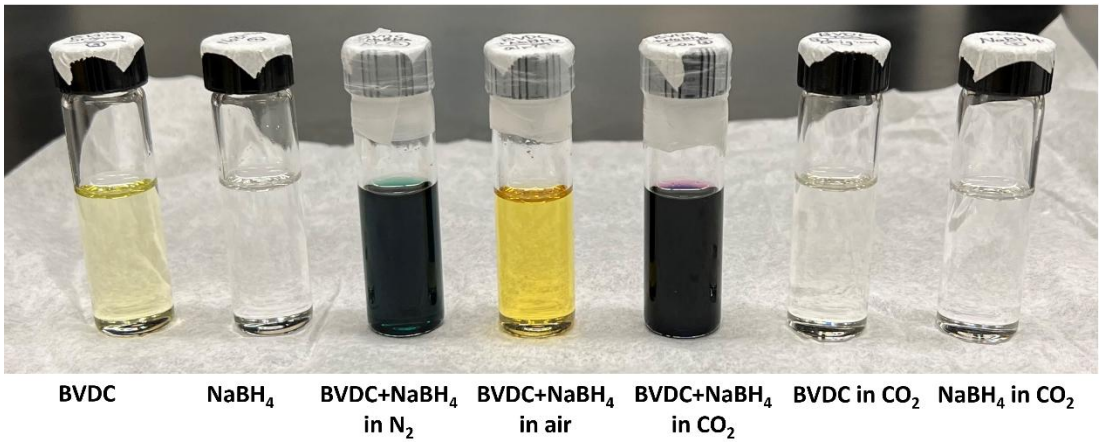

Figure S10 | The demonstration experiment on the interactions between reduced viologen structures and CO_2 molecules.

As shown in Figure S10, the colour of the viologen solution changed from light yellow to dark green, owing to the reduction reactions. When the reduced solution was with air, the reduced viologen quickly turned yellow, indicating that the reduced viologen was oxidized by O_2 in the air. Instead, when we purged CO_2 in the reduced solution, the colour changed to purple or blue, which demonstrated the possibility that CO_2 was bonded to the reduced viologen structure. The transfer of electrons from the reduced viologen to the bonded CO_2 molecules caused the colour change. This experiment verifies the assertion that the reduced viologen species have the capability to activate CO_2 molecules.

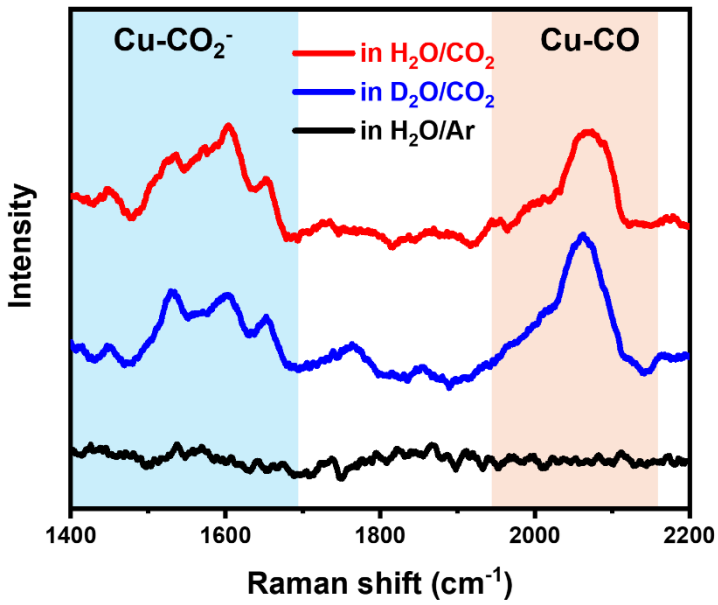

Figure S11 | The Raman spectra of the PTV-Cu electrode scanning at -0.43 V vs. RHE in H_2O/CO_2, D_2O/CO_2 and H_2O/Ar atmosphere.

As shown in Figure S11, the Raman spectrum in D_2O/CO_2 displays a similar curve to the one in H_2O/CO_2, while the peaks around 1600 cm^{-1} disappear in the H_2O/Ar atmosphere, indicating that the corresponding peaks should be assigned to CO_2^-.

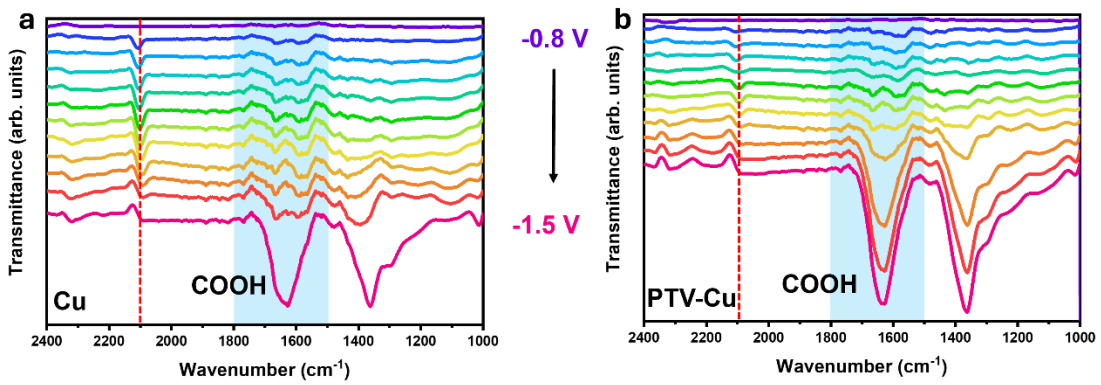

Figure S12 | Corresponding In-situ FTIR spectra of a. bare Cu electrode and b. PTV-Cu electrode scanning from -0.8 to -1.5 V vs. Ag/AgCl from Figure 2c-d.

The peak at 2100 cm^{-1} is initially absent when a potential of -0.8 V is applied, and it increases in intensity as the applied potential becomes more negative. Here we discuss whether this is significant, e.g. is it meaningfully correlated with *CO. The peak at 2100 cm^{-1} is consistently a downward feature across studies (Figure S12), a finding that is consistent with an increase in *CO coverage at more negative potentials.[1-2] The upward peak adjacent to the *CO peak, particularly at more negative potentials, may tentatively be ascribed to the consumption of CO_2 during the reaction. The shift in the peak position may be attributed to the different electrolytes used (1 M K_2CO_3 instead of 1M KOH or 1 M $KHCO_3$), wherein variation in pH and/or ionic strength may account for a minor shift compared to references.[1-3]

Compared to the Raman spectra in Figures 2a and 2b, the intensity of the FTIR spectra is relatively weak. This is because Cu substrates exhibit excellent Raman enhancement in the aqueous media, while the high sensitivity of the IR light towards water molecules decreases the signal-to-noise ratios, leading to the generally weaker *CO signals on the catalysts.[1-3]

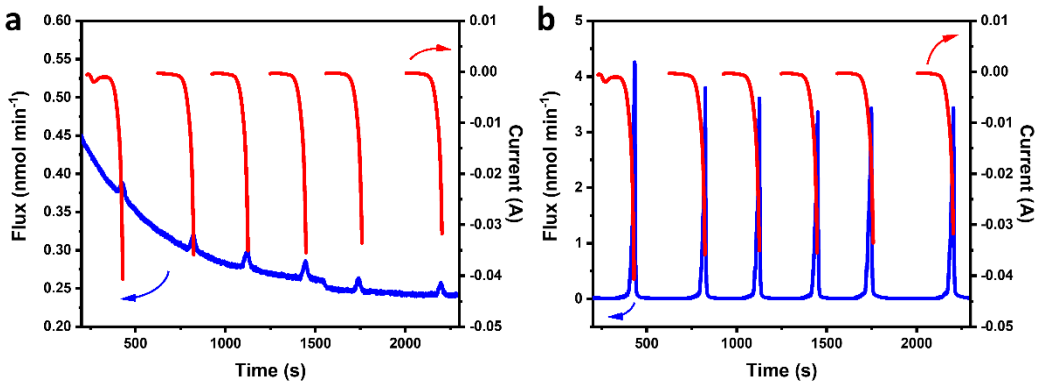

Figure S13 | in-situ DEMS results of a. C_2H_4 evolution and b. H_2 evolution with PTV-Cu electrode and the corresponding LSV tests for 6 cycles.

The results in Figure S13 show that both H_2 and C_2H_4 evolution at different cycles occurs at the same potential with similar intensity ratios, verifying that the addition of PTV can indeed facilitate CO2RR while suppressing HER by activating CO_2 molecules.

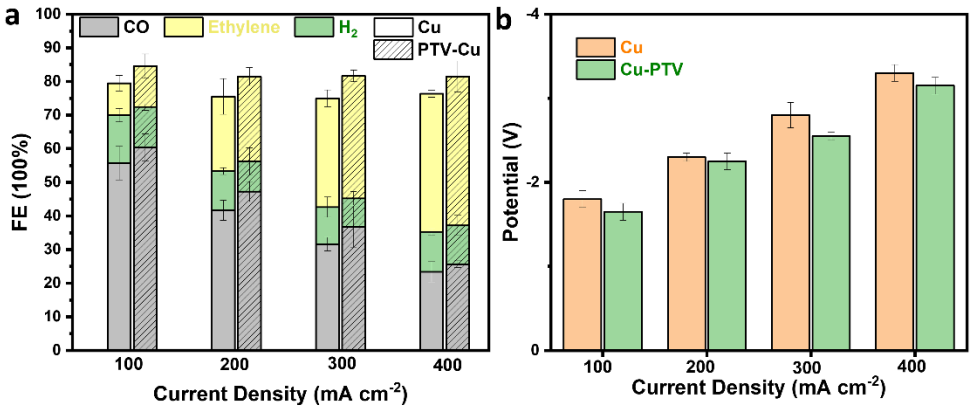

Figure S14 | Electrochemical performance of CO2RR when bare Cu electrode and PTV-Cu electrodes are used in a three-electrode cell. a. The product distribution for the electrochemical conversion of CO_2 at current densities from 100 to 400 mA cm^{-2}. **b.** The corresponding cathodic potentials of the flow cell when bare Cu and PTV-Cu electrodes are used.

The addition of PTV in the electrode slightly changed the product distribution towards carbon products instead of H_2. However, the potentials of PTV-Cu dropped compared to bare Cu. These results indicate that PTV can promote earlier CO_2 conversion by activating CO_2 molecules at a lower potential while facilitating CO2RR to suppress HER. However, its capability to change the product distribution is limited as it performs better in a CO_2 lean condition, where PTV can trap CO_2 while initiating the CO2RR by activating CO_2 molecules.

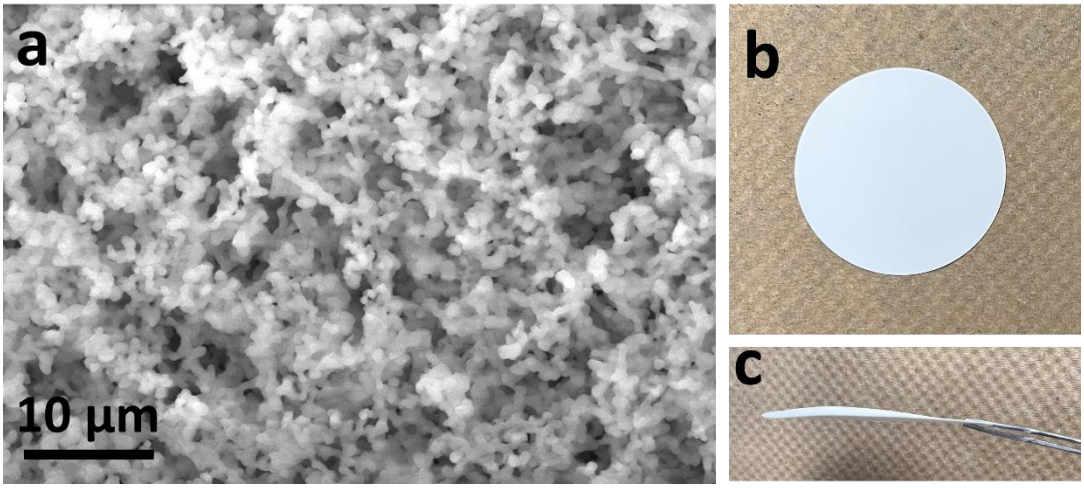

Figure S15 | The a. morphology and b-c. digital photos of the MCE buffer layer. The commercial MCE filter paper contains porous structures with around 8 µm pores and is composed of small-sized microparticles.

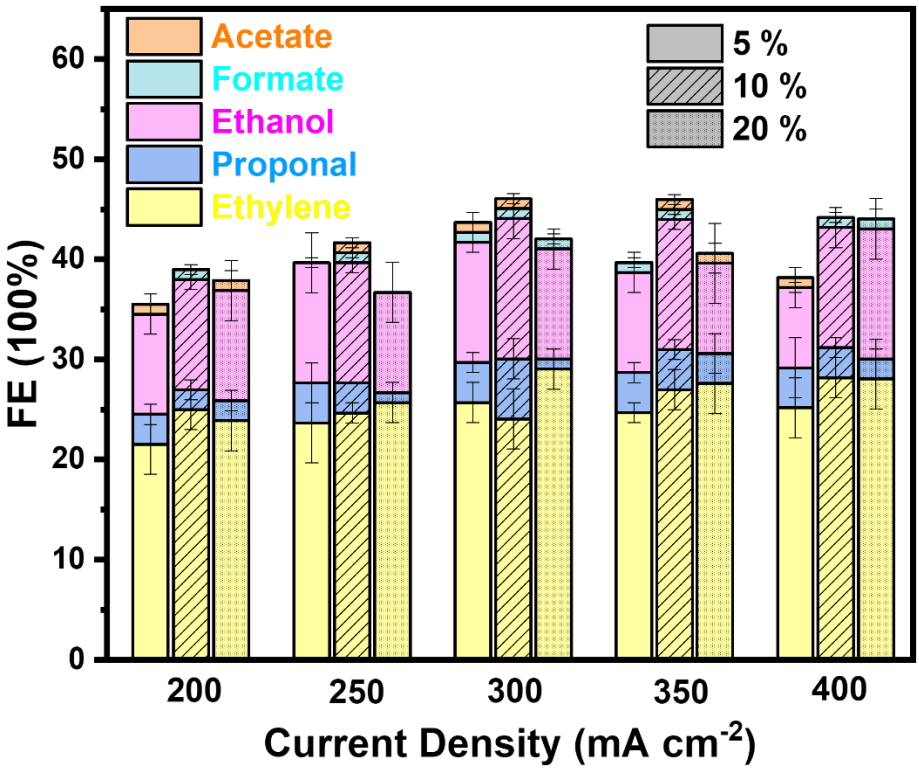

Figure S16 | The product distribution for the electrochemical conversion of 1 M K_2CO_3 at current densities from 200 to 400 mA cm^{-2} with different PTV loading.

The increasing of the PTV loading in the electrode increases the production of C_{2+} products, owing to the capability of activating CO_2. However, after the loading amount exceeds 10 %, it may start to block access to CO_2 molecules of the Cu nanoparticles. Since PTV can trap CO_2 molecules in the molecular structures, too much PTV loading may increase the transfer distance between the trapped CO_2 molecules and the Cu surface, which further leads to the decrease in the electrochemical performance.

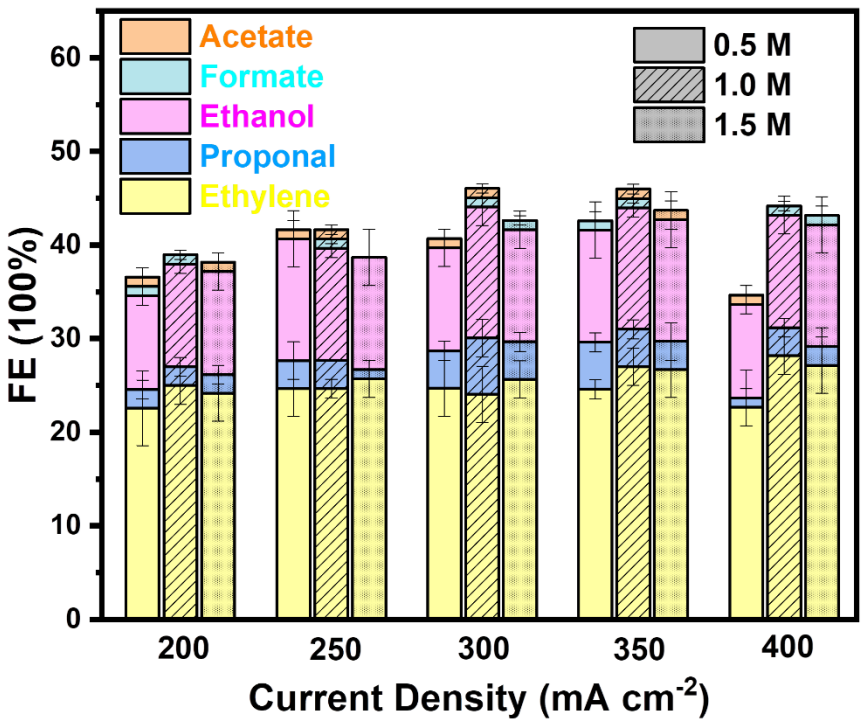

Figure S17 | The product distribution for the electrochemical conversion of K_2CO_3 with different concentrations at current densities from 200 to 400 mA cm^{-2} with PTV-Cu electrode.

Increasing carbonate concentration provides more CO_2 sources, higher pH around the electrode, and higher ionic conductivity. There is not much change when further increasing the concentration over 1 M, since the CO_2 generation rate is limited by proton generation on the membrane.

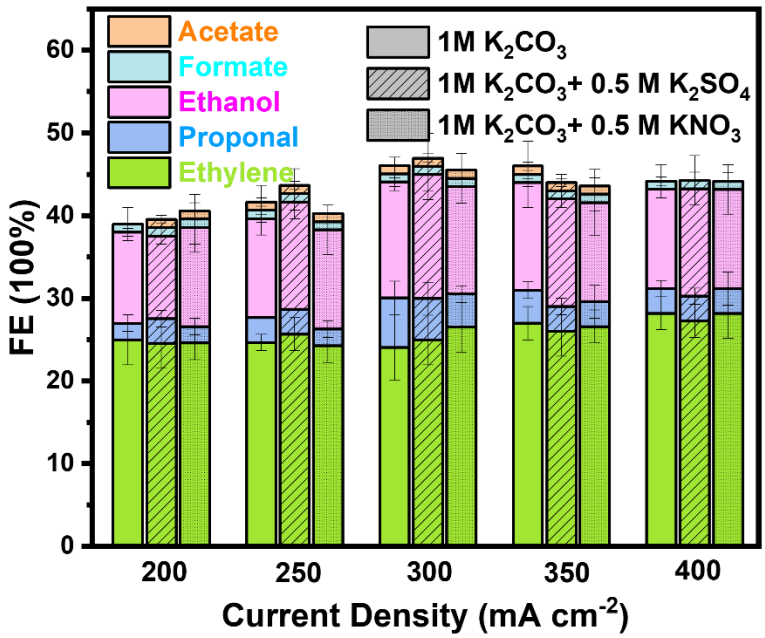

Figure S18 | The product distribution for the electrochemical conversion of 1 M K_2CO_3, 1 M K_2CO_3/0.5 M K_2SO_4, and 1M K_2CO_3/0.5 M KNO_3 at current densities from 200 to 400 mA cm^{-2} with PTV-Cu electrode.

We further explore the possibility of this reactive-capture system using flue gas. Since SO_2 and NO_x are readily absorbed by OH^- to form SO_4^{2-} and NO_3^- in the presence of air, we simulated this scenario by introducing these compounds into a 1 M K_2CO_3 electrolyte. As shown in Figure S18, the FE_{C2+} remains largely unchanged across different composition variations, suggesting that impurity gases may not significantly impact the electrified reactive capture process. This is likely because the process relies on the protonation of CO_3^{2-} to produce CO_2 in situ for conversion, while the impurity ions are less likely to release gas at low pH and concentrations. Additionally, the ability of PTV to enrich the local CO_2 concentration and activate CO_2 may further mitigate the negative effects, making it a promising candidate for future industrial applications.

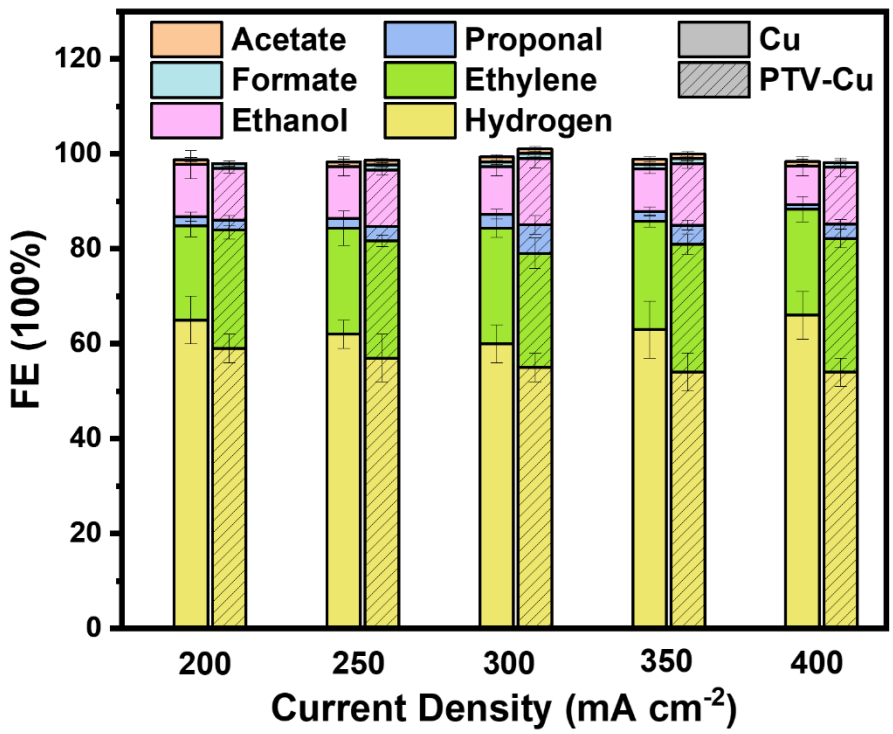

Figure S19 | Complete product distribution for the electrochemical conversion of 1 M K_2CO_3 solution at current densities from 200 to 400 mA cm^{-2} using bare Cu and PTV-Cu electrodes shown in Figure 3b.

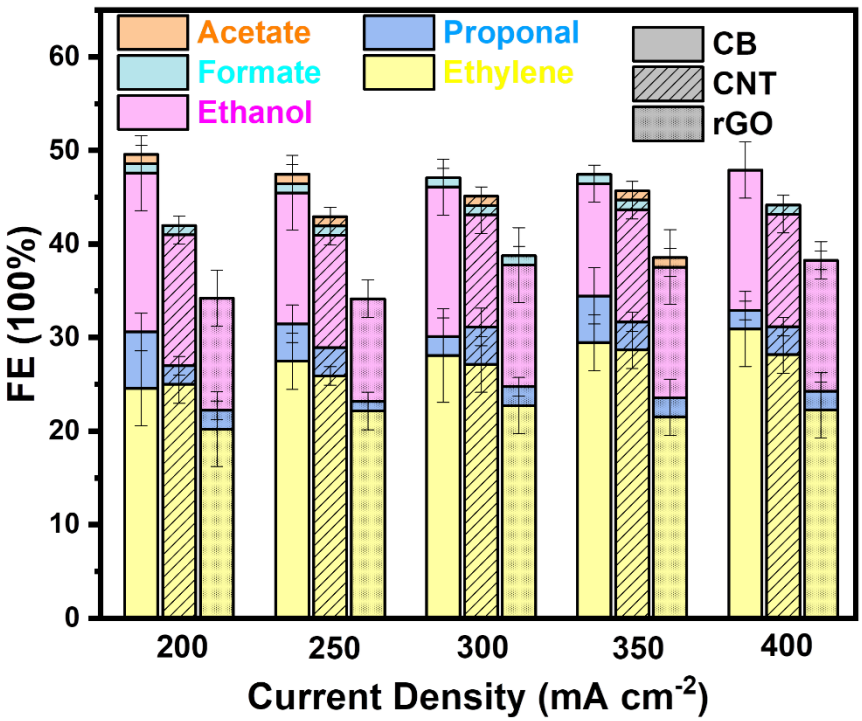

Figure S20 | The product distribution for the electrochemical conversion of 1 M K_2CO_3 at current densities from 200 to 400 mA cm^{-2} with different carbon fillers such as CB, CNT and rGO.

Both CB and CNT perform well with PTV because they increase the overall conductivity which is beneficial for redox efficiency, which can further increase the C_{2+} selectivity while reducing the local H_2O content, leading to reduced FE_{H2} and increased FE_{C2+}. CB performs the best of the two due to a higher surface area and better distribution in the electrode. rGO is also conductive, but its sheet structure may prevent the transfer of activated CO_2 molecules to the surface of the Cu electrode, thus leading to lower C_{2+} production.

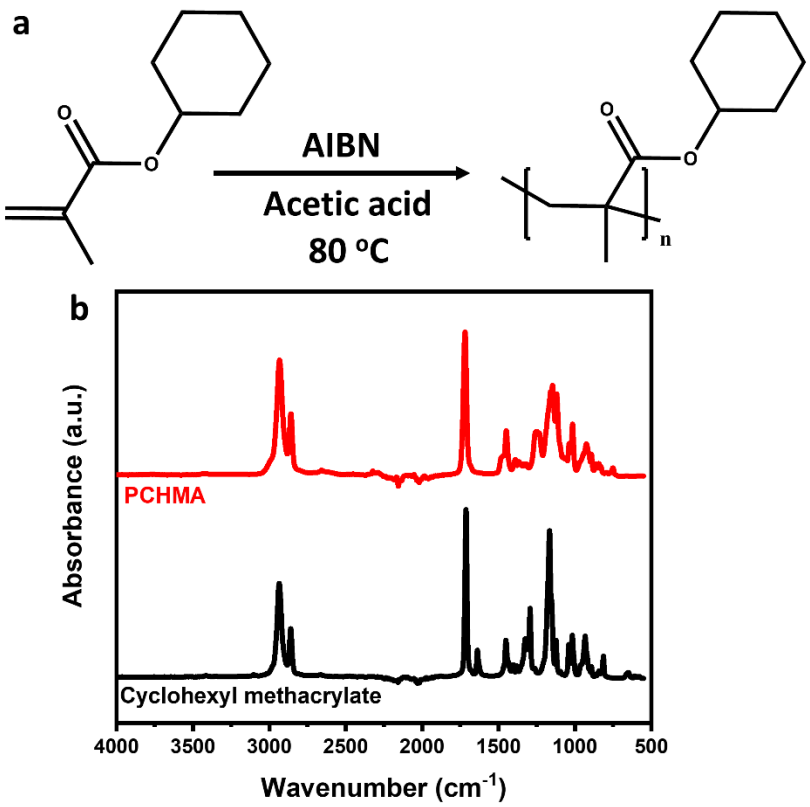

Figure S21 | a. The schematic illustration of PCHMA synthesis and b. the characterizations of as-synthesized PCHMA and precursors.

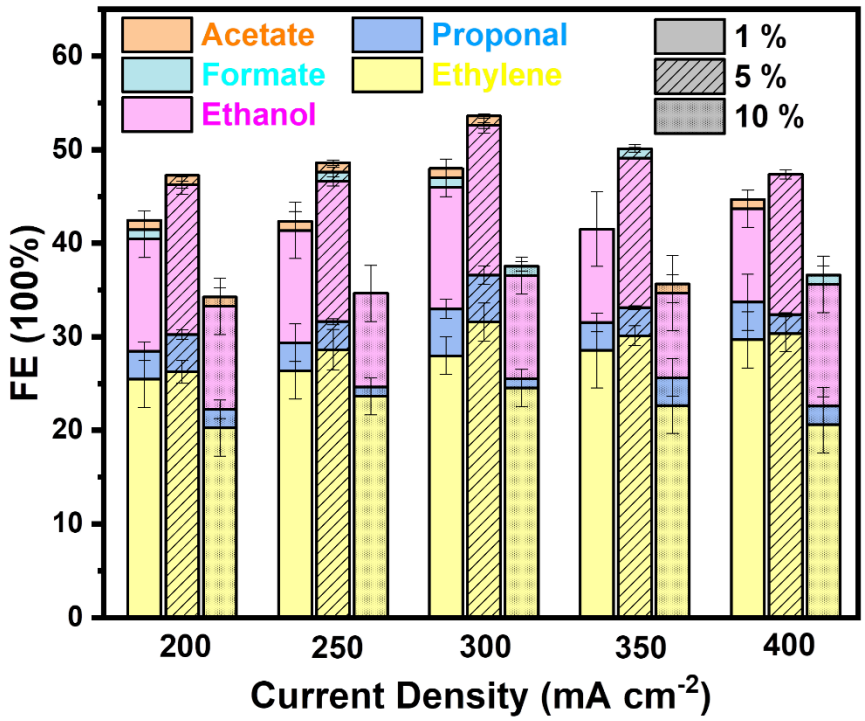

Figure S22 | The product distribution for the electrochemical conversion of 1 M K_2CO_3 at current densities from 200 to 400 mA cm^{-2} with different PCHMA loading on the PTV-Cu electrode.

The addition of the PCHMA layer on the PTV-Cu electrode can improve C_{2+} production because PCHMA is hydrophobic, which can reduce the surface water content. However, too much PCHMA loading over 5 % leads to the blockage of ions and CO_2 transfer. Thus 5% is the most optimized loading of PCHMA on the electrode

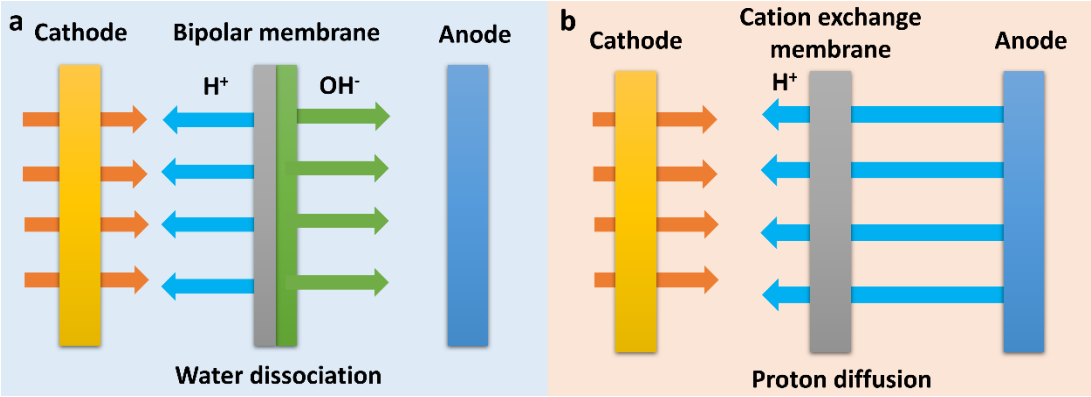

Figure S23 | The schematic illustrations of the functional mechanisms when a. a BPM with alkaline anolyte and b. CEM with acidic anolyte is used.

The in-situ generation of CO_2 molecules relies on the generation of protons on the electrode which further react with carbonate in the catholyte. The most commonly used membrane in the system is BPM. However, the water dissociation process increases the overall voltage, leading to energy loss in the system. The replacement of BPM with an alkaline anolyte with CEM with an acidic anolyte may help lower the potential, especially the change of anode catalyst from Ni foam in an alkaline condition to IrO_x/Ti in an acidic condition, which can also help bring down the full cell voltage.

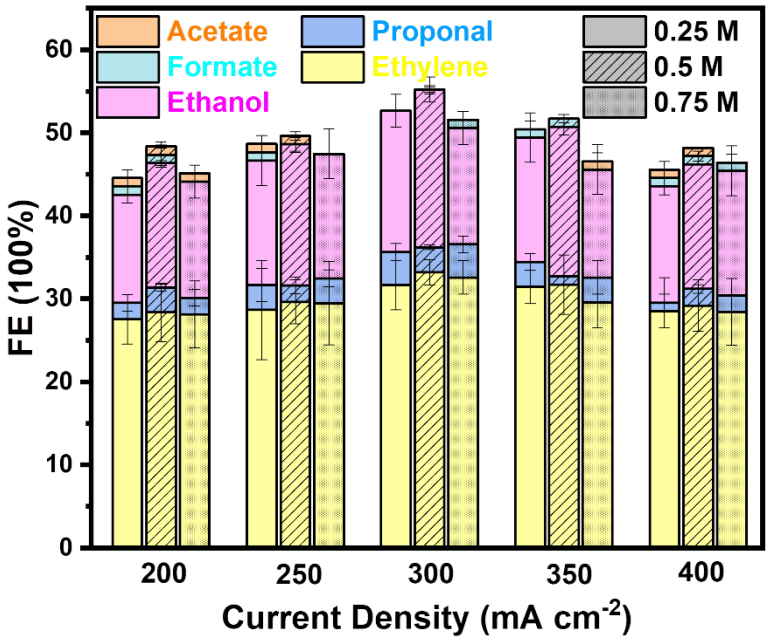

Figure S24 | The product distribution for the electrochemical conversion of 1 M K_2CO_3 at current densities from 200 to 400 mA cm^{-2} with PCHMA-PTV/CB-Cu electrode when H_2SO_4 with different concentrations are used as anolyte.

The concentration of H_2SO_4 does not have a significant influence on the electrochemical performance at the catholyte.

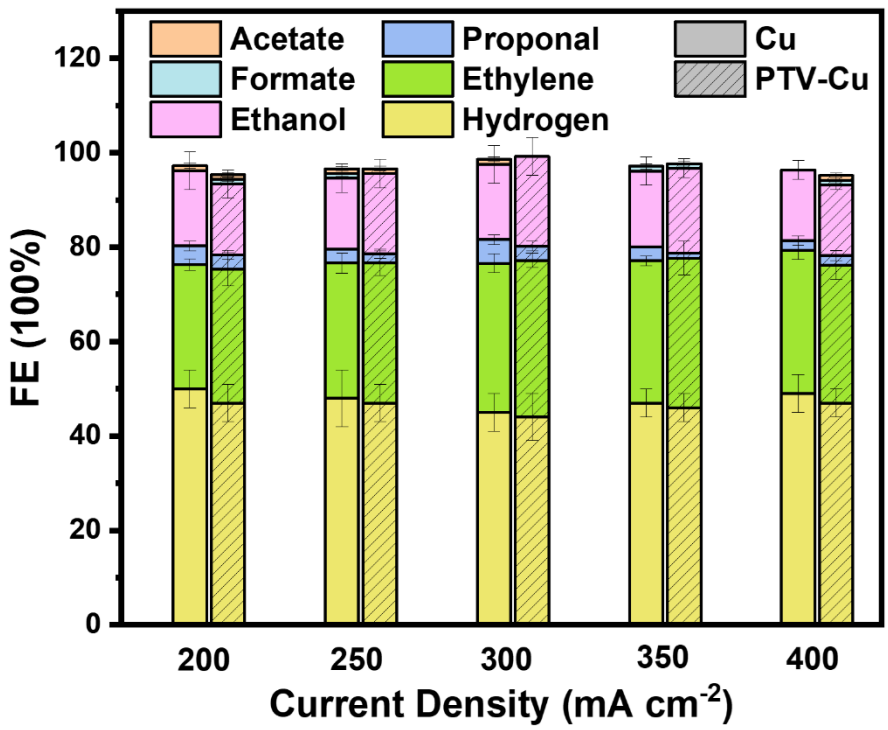

Figure S25 | Complete product distribution for the electrochemical conversion of 1 M K_2CO_3 solution at current densities from 200 to 400 mA cm^{-2} with PCHMA-PTV/CB-Cu electrode with alkaline anolyte with BPM and acidic anolyte with CEM, respectively, shown in Figure 3d.

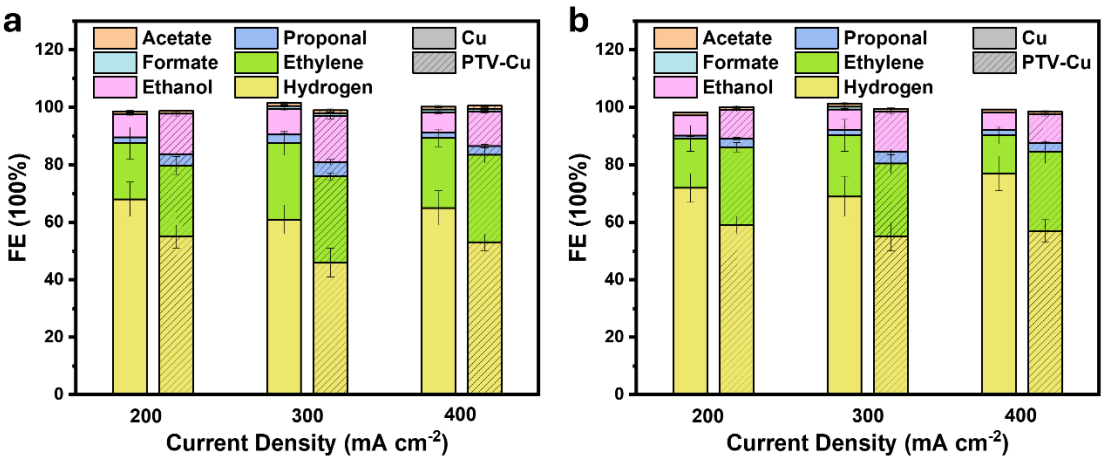

Figure S26 | Complete product distribution for the electrochemical conversion of the captured liquids at current densities from 200 to 400 mA cm^{-2} using bare Cu and PTV-Cu electrodes, shown in Figure 4a, capturing a. 100% CO_2 and b. 1% CO_2.

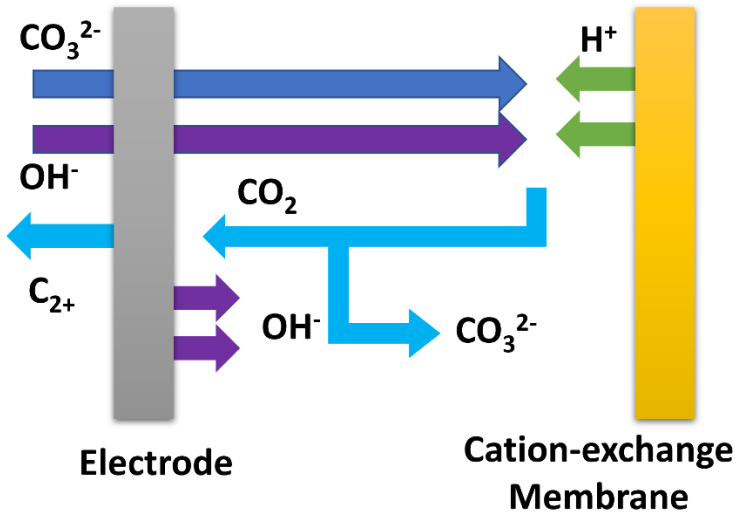

Figure S27 | The schematic illustration of the reactions that occurred in the capture liquid.

The high concentration of OH^- from the mixture solution continuously exhausts the protons from the membrane, and the amount of CO_2 molecules generated by the reactions between the remaining protons and carbonate is very limited.

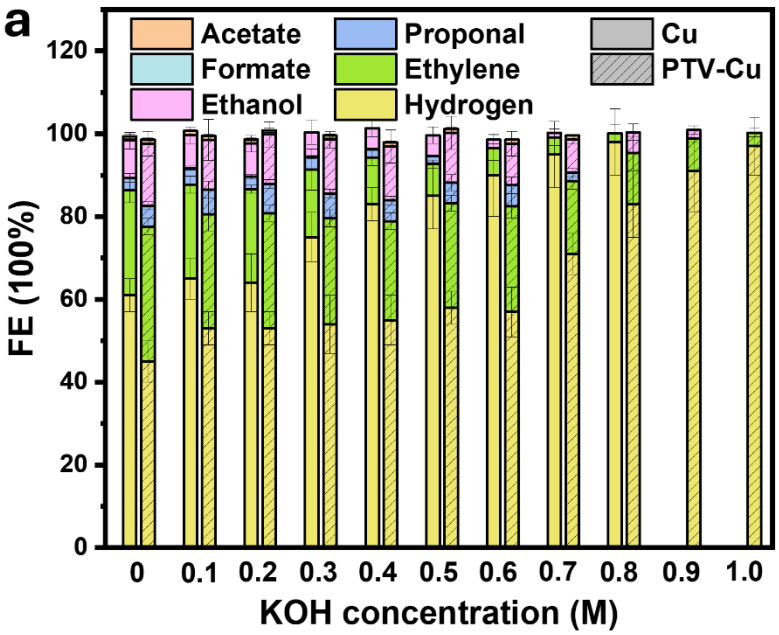

Figure S28 | The complete product distribution for the electrochemical conversion of mixture liquids at the current densities of 300 mA cm^{-2} using bare Cu and PTV-Cu electrodes, shown in Figure 4b.

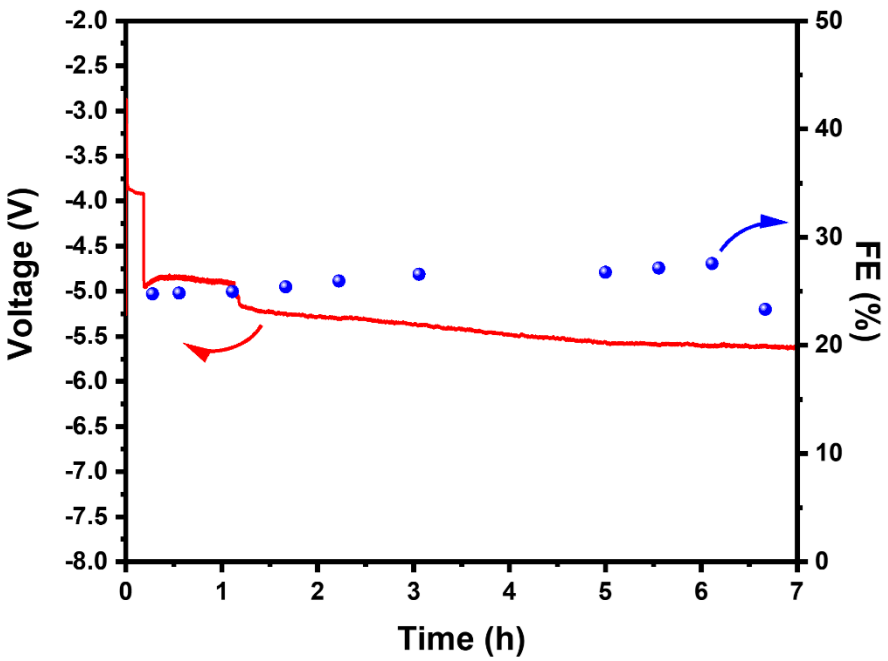

Figure S29 | The Faradaic efficiencies of C_2H_4 products, and cell voltage for the stability test with PCHMA-PTV/CB-Cu electrode and MCE buffer layer in alkaline condition. The current density was 300 mA cm^{-2}.

The stability test was carried out using the optimized PCHMA-PTV/CB-Cu electrode and MCE buffer layer. We used 1 L 1 M K_2CO_3 as the catholyte and 1 L 1 M KOH as the anolyte. We chose BPM as the membrane as it could better suppress the product cross-over than CEM. We found that the cell could only last around 6 h, and the voltage continuously decreased during the test. We assigned the failure mechanism to the increase of pH and destruction of the MCE filter layer.

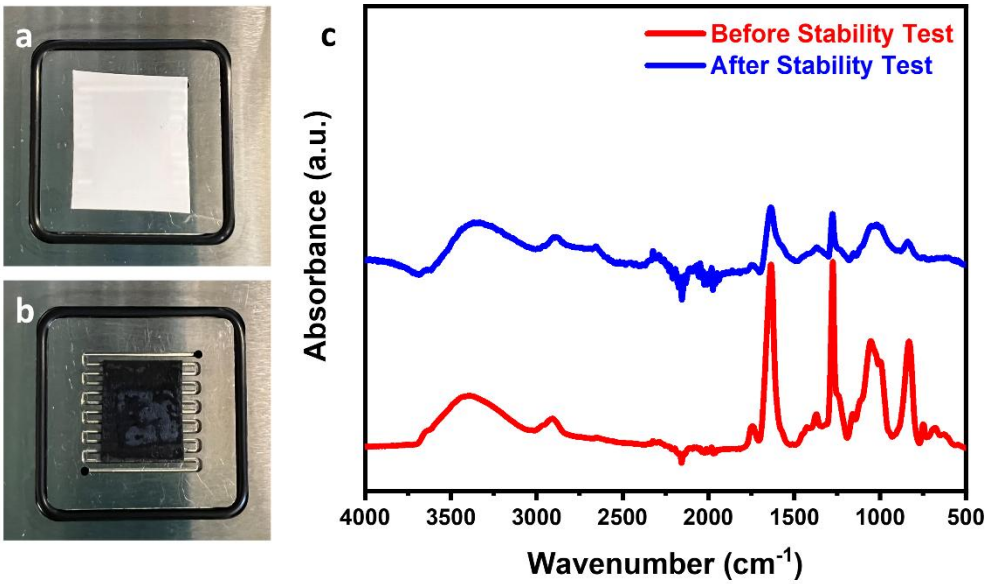

Figure S30 | The characterization of the PTV electrode before and after stability tests. **a-b.** The digital photo of the electrode and MCE filter paper **a.** before and **b.** after stability tests. **c.** The FTIR spectra of the electrode and filter paper after the stability tests.

The structure of MCE filter paper was destroyed after long-term stability tests (Figure S30a-b). Moreover, the decomposed filter fragments remain on the surface of the electrode, which may block the active sites on the electrode, affecting the overall electrochemical performance.

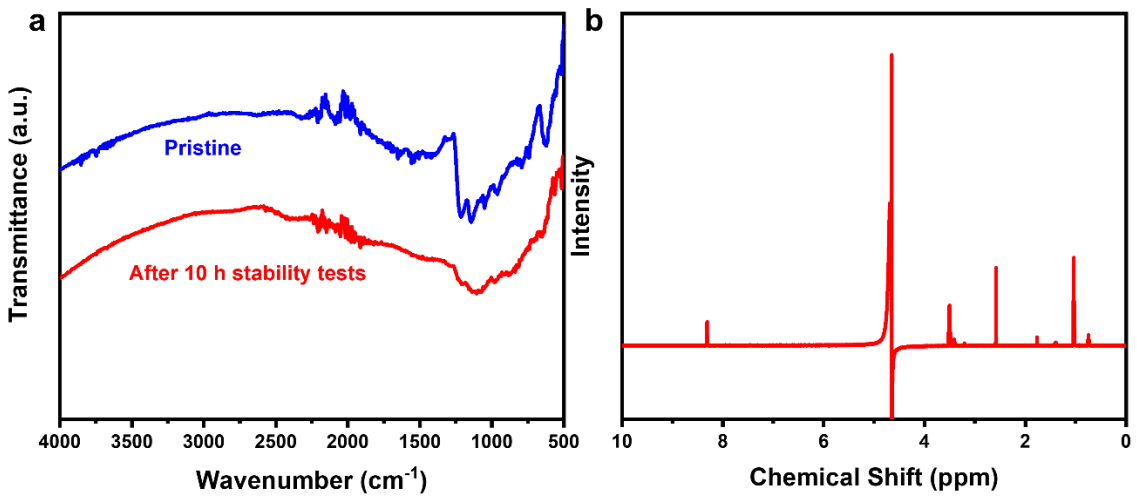

Figure S31 | FTIR spectra of the PTV-Cu electrode before and after stability tests.

To assess the stability of PTV during the electrochemical process, we performed FTIR analysis on the PTV-Cu electrode both before and after conducting electrified reactive capture tests. The electrode was soaked in water with mild shaking to remove the surface MCE filter paper residue. As shown in Figure S31a, the electrode exhibits negligible changes after testing compared to the pristine electrode, confirming that PTV remains stable throughout the electrocatalysis. Additionally, we conducted NMR measurements of the electrolyte post-test, and Figure S31b shows no evidence of pyridyl-based compounds. These results confirm that PTV remains structurally stable and does not decompose into pyridyl-based compounds during the process.

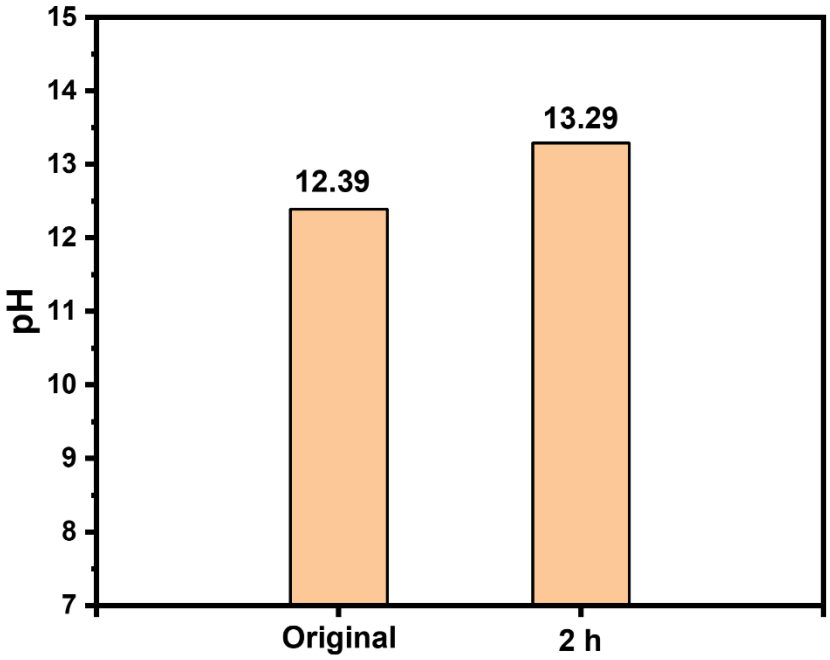

Figure S32 | pH change in the catholyte during the electrochemical test after 2h. The current density was 300 mA cm^{-2}.

Owing to the generation of OH^- during HER and CO2RR, the pH of the catholyte quickly climbed up from 12 to over 13 in 2 h. This could explain the performance drop during long-time stability tests and the destruction of the MCE structure.

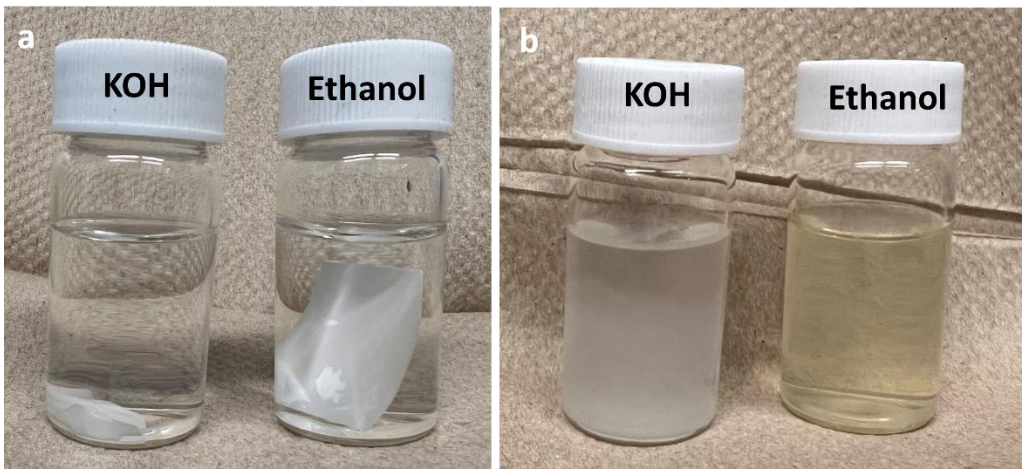

Figure S33 | Digital photos of the MCE filter paper a. before and b. after soaking in KOH and ethanol.

As shown in Figure S33, the structure of the MCE filter paper was destroyed after soaking in KOH and ethanol, since the structure is integrated mainly by ether bonds. This indicates that the MCE filter paper is not stable in strong alkaline conditions and high-concentrated ethanol solutions. Therefore, a new resistant buffer layer mimicking the structure of MCE should be developed for long-term stability tests.

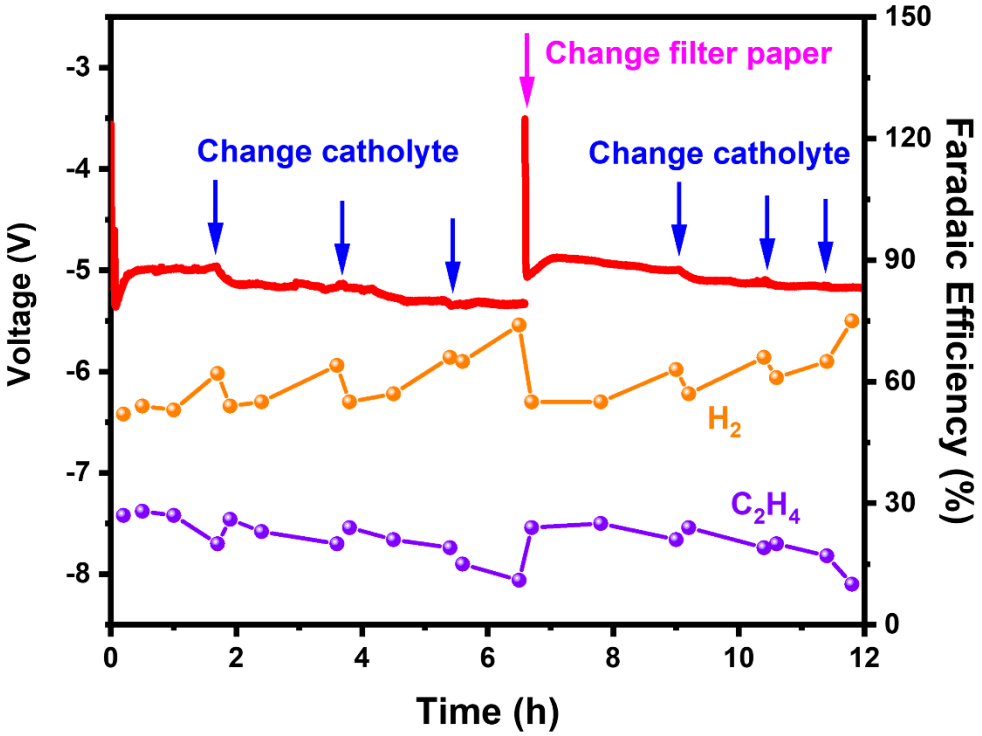

Figure S34 | Demonstration experiment on the influence of pH and filter paper on the electrochemical stability performance of carbonate reduction. Bare Cu electrode was used for the test. The current density was kept at 300 mA cm^{-2}. The catholyte was changed during the test when the FE_{C2H4} started to drop.

The FE_{C2H4} recovered when the catholyte was changed with fresh ones in the first two times, indicating in the early stage of the stability test, the increase of pH and probable product accumulation may be the limiting factor for the stability test. However, the FE_{C2H4} after the third time change of catholyte could not recover, and only the change of MCE buffer layer could increase the FE_{C2H4}. This verifies that after longer period of tests, the MCE filter paper decomposed, leading to the destruction of the porous buffer layer which could alter the pH gradient and limit CO_2 transport during tests.

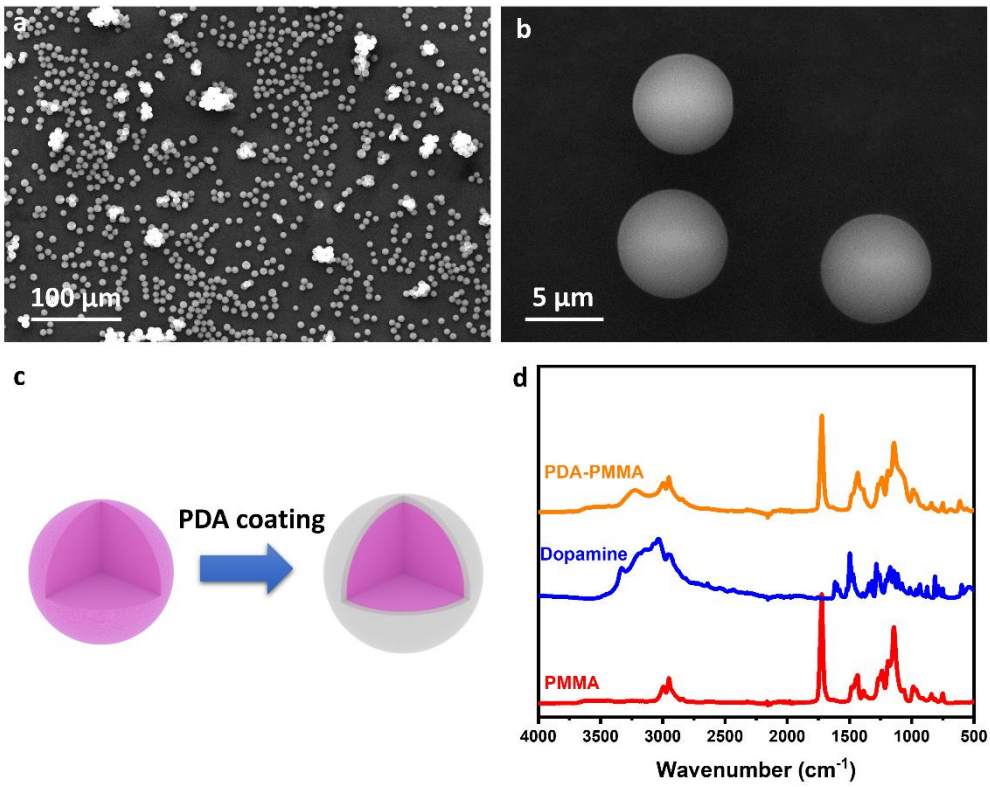

Figure S35 | The characterizations of the PDA-PMMA microspheres. The SEM images of the synthesized PMMA microspheres at **a.** low and **b.** high magnifications. **c.** The schematic illustration of coating PDA on PMMA microspheres. **d.** The FTIR spectra of the PMMA microspheres, dopamine precursor, and the PDA-PMMA microspheres.

The coating of PDA on the surface of PMMA microspheres did not change the morphology. The coating of hydrophilic PDA on hydrophobic PMMA microspheres makes the PDA-PMMA hydrophilic, which will allow the carbonate solution to access the buffer layer.

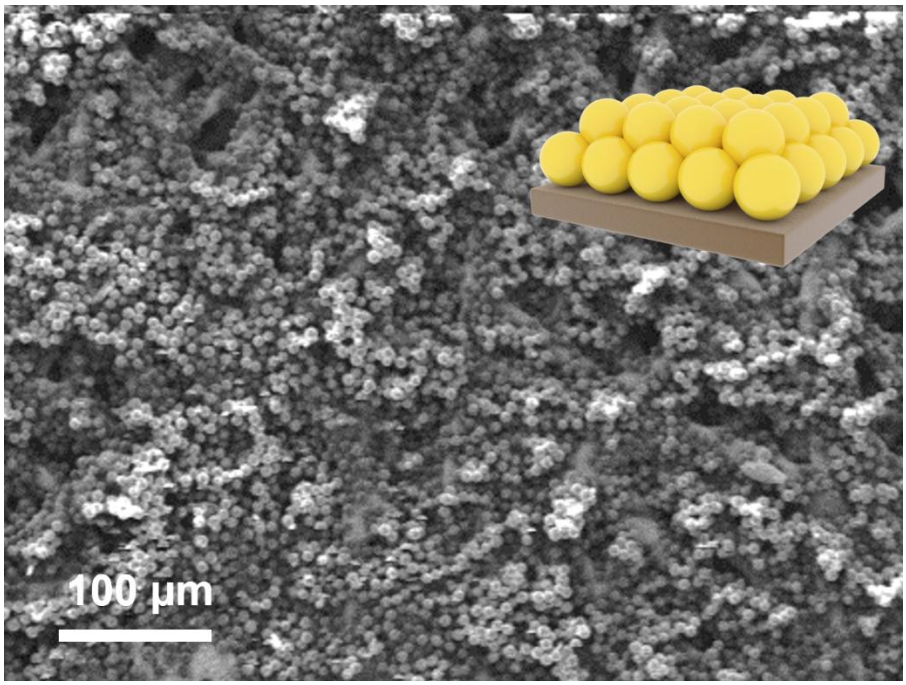

Figure S36 | SEM image of the electrode surface with the Nafion/PDA-PMMA buffer layer. The inset is the schematic illustration of the Nafion/PDA-PMMA microspheres dispersed on the electrode.

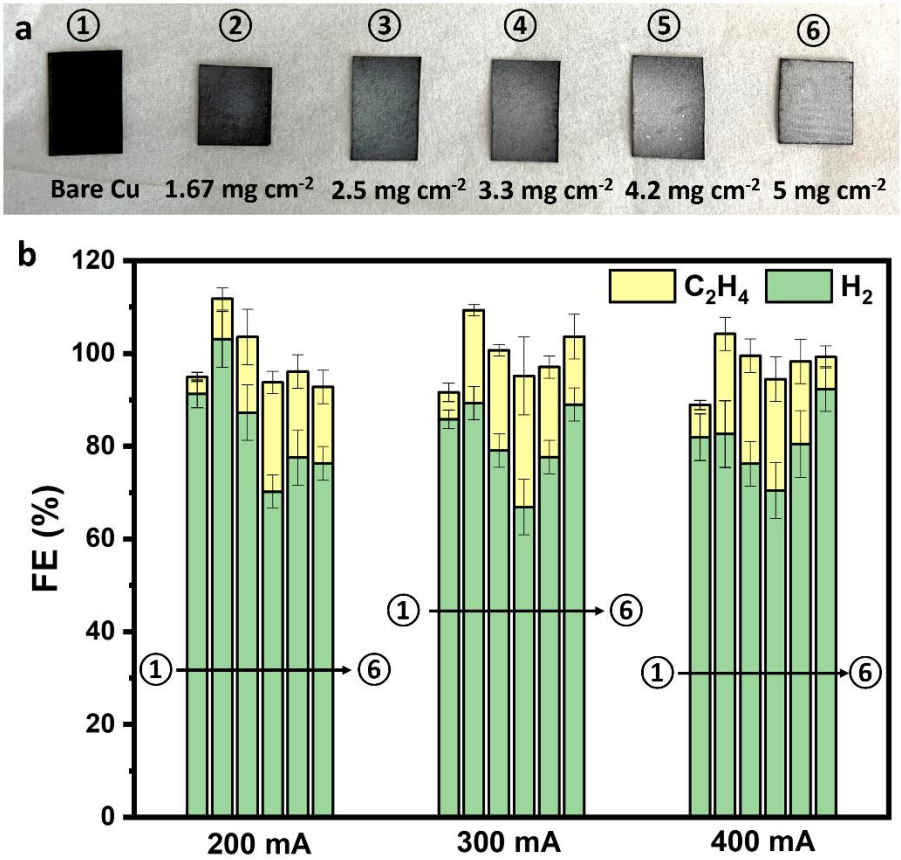

Figure S37 | Loading of PDA-PMMA microspheres on electrodes. a. The digital photo of the electrodes after spraying different amounts of PDA-PMMA microspheres with Nafion. **b.** The product distribution when electrodes with different loading amounts are used. Only H_2 and C_2H_4 are tracked for this experiment. The electrodes used here are the bare Cu electrode. The catholyte is 1 M K_2CO_3, and the anolyte is 1 M KOH. BPM is used to separate anolyte and catholyte.

We explored the loading effect of the PDA-PMMA microspheres on bare Cu electrodes. The increase in the loading promotes the generation of C_2H_4 while suppressing HER, because a higher loading amount creates a thicker buffer layer which will allow an optimized pH gradient layer to generate CO_2 and convert CO_2 to C_{2+} products. A thinner buffer layer can result in low pH on the electrode surface, which produces more H_2. However, when the loading amount exceeds 3.3 mg cm^{-2}, the pathway of CO_2 transfer becomes too long and insufficient CO_2 access on the electrode, which results in the increase of HER.

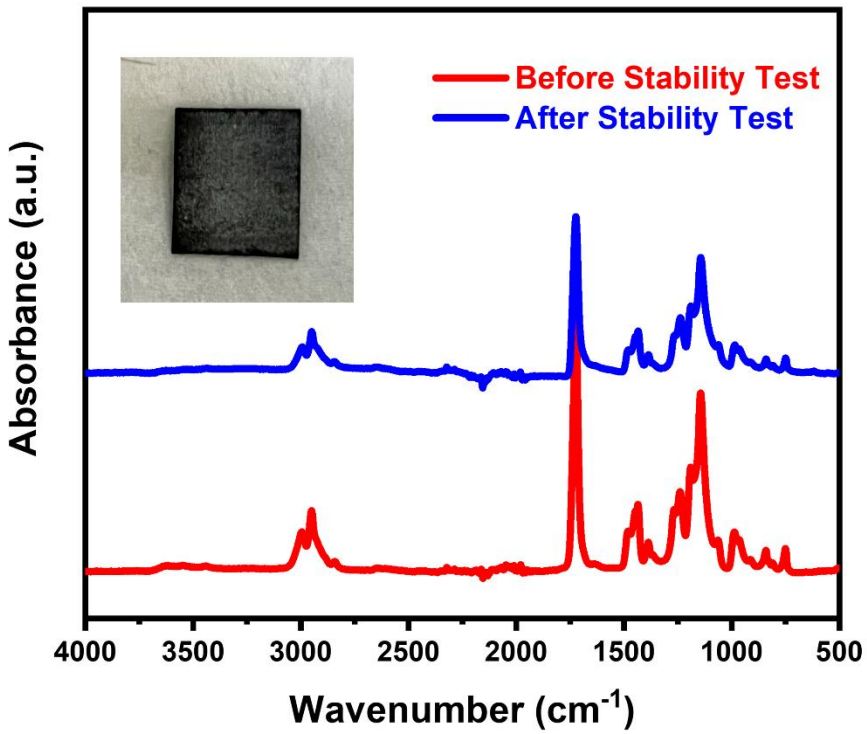

Figure S38 | FTIR spectra of the electrode before and after the stability test.

There is no visible change in the spectra, indicating no change of the buffer layer occurs during the stability test.

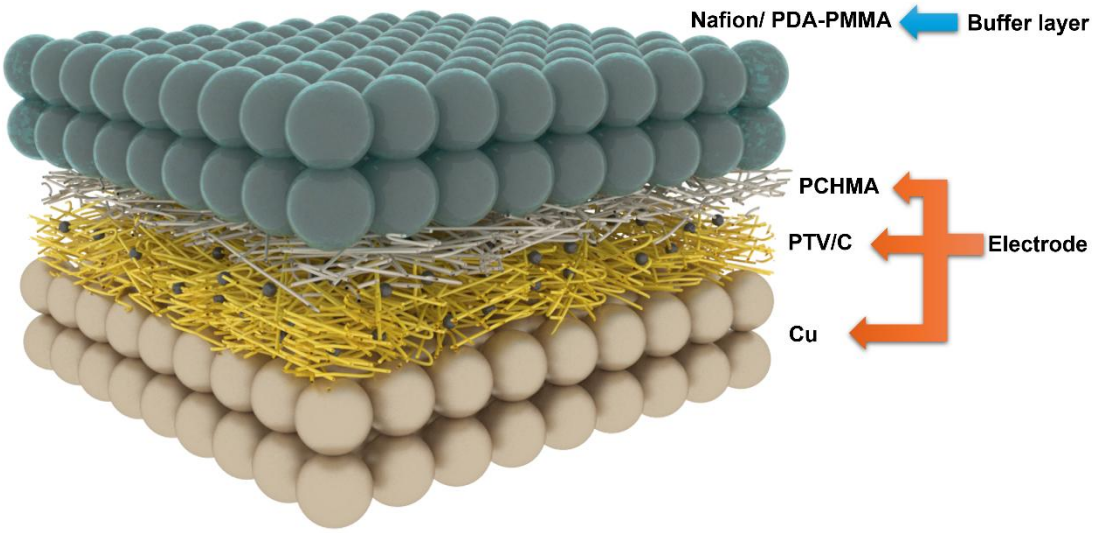

Figure S39 | The schematic structure of the optimized electrode with the top-loaded buffer layer for the integrated reactive capture operation.

The electrode consists of three functional layers: a Cu layer at the bottom to convert activated CO_2 into C_{2+} products; a middle PTV/C layer for trapping and activating CO_2 to enhance reaction kinetics; and a hydrophobic PCHMA layer on top to reduce water content at the electrode surface. The optimization of these layers has been clearly illustrated in Figures S16 and S20-22.

We have also developed an improved interposer layer (Nafion/PDA-PMMA buffer layer) that replaces the MCE interposer layer employed in prior studies (Figures S35-38). The buffer layer shows much-improved stability in alkaline and alcohol conditions than the MCE interposer layer.

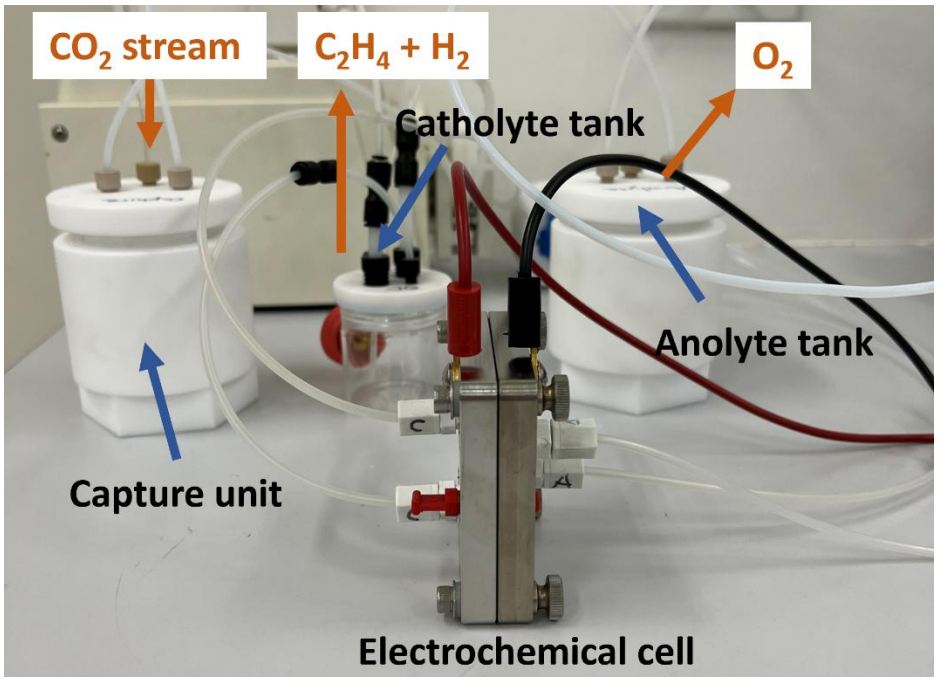

Figure S40 | The schematic illustration of the re-designed integrated system combining CO_2 capture and conversion.

The integrated system consists of four major sections: capture tank, catholyte tank, electrochemical cell, and anolyte tank. The anolyte tank has 1 M KOH which circulates into the anodic flow chamber for anodic reactions. The catholyte tank contains 1 M K_2CO_3 which circulates into the cathodic flow chamber for the conversion reactions. The catholyte flows into the capture tank for capturing CO_2 by utilizing the OH^- generated from the electrochemical cell. Thus as a result, the system continuously captures CO_2 into K_2CO_3 which is used as the carbon feedstock for the conversion into C_{2+} products.

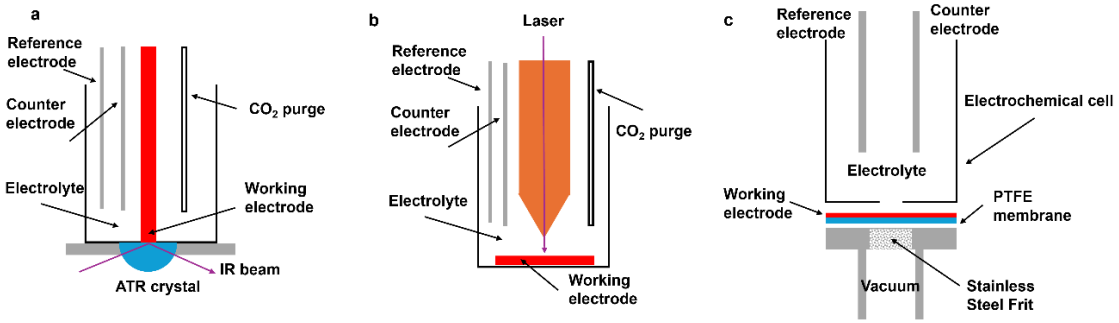

Figure S41 | The schematic illustration of the homemade in-situ cells for a. FTIR, b. Raman, and c. DEMS.

Table S1 | Computed reduction potential of different species

Process	Computed reduction potential (V *vs.* NHE)
$CO_2 + e^- \rightarrow CO_2^-$	-2.00
$PTV^{2+} + e^- \rightarrow PTV^{\bullet+}$	-0.52
$PTV^{\bullet+} + e^- \rightarrow PTV^{(0)}$	-0.99
$PTV^{(0)}\text{-}CO_2 + e^- \rightarrow [PTV^{(0)}\text{-}CO_2]^{\bullet-}$	-1.50

Table S2 | Energy analysis of cost in different systems, alkaline CO_2RR in a flow cell, neutral CO_2RR in a membrane electrode assembly (MEA) cell and carbonate electrolysis.

System		Alkaline CO_2RR	Neutral CO_2RR	Carbonate electrolysis (BPM)	Carbonate electrolysis (CEM)
Full cell voltage (V)		2.4	3.9	4.9	4.1
C_2H_4 selectivity (%)		70	66	32	33
Current density (mA cm^{-2})		150	315	300	300
Energy efficiency (%)		34	19	8	9
CO_2 utilization (%)		5	11	100	100
C_2H_4 concentration at the outlet (wt.%)		4	8	55	56
CO_2 concentration at the outlet (wt.%)		93	88	0	0
Energy cost (GJ/tonne of C_2H_4)					
Upstream generation		28	28	3	3
Electrolysis		142	244	633	514
Product separation	CO_2 removal	88	42.1	0	0
	Dehydration	1.4	0.9	0.8	0.8
	Distillation	26	12	1.4	1.4
Anode separation		0	57	0	0
Carbonate regeneration		198	0	0	0
Total		483	384	638	519

Supplementary References:

1 Li, X. *et al.* Selective visible-light-driven photocatalytic CO_2 reduction to CH4 mediated by atomically thin CuIn5S8 layers. *Nature Energy* **4**, 690-699 (2019).

2 Zhang, Q. *et al.* Designing covalent organic frameworks with Co-O4 atomic sites for efficient CO_2 photoreduction. *Nature Communications* **14**, 1147 (2023).

3 Bagchi, D. *et al.* Structure-tailored surface oxide on Cu–Ga intermetallics enhances CO_2 reduction selectivity to methanol at ultralow potential. *Advanced Materials* **34**, 2109426 (2022).